Chapter 9 Problems

The first section of problems belongs to the basic category, and the answers are provided in the back of the book. The remaining three sections contain problems belonging to the basic, advanced, and extension categories, respectively.

BASIC PROBLEMS WITH ANSWERS

9.1. For each of the following integrals, specify the values of the real parameter σ which ensure that the integral converges:

(a) $\int_0^\infty e^{-5t}e^{-(\sigma+j\omega)t}\,dt$ **(b)** $\int_{-\infty}^0 e^{5t}e^{-(\sigma+j\omega)t}\,dt$

(c) $\int_5^5 e^{5t}e^{-(\sigma+j\omega)t}\,dt$ **(d)** $\int_{-\infty}^{\infty} e^{-5t}e^{-(\sigma+j\omega)t}\,dt$

(e) $\int_{-\infty}^{\infty} e^{-5|t|}e^{-(\sigma+j\omega)t}\,dt$ **(f)** $\int_{-\infty}^{0} e^{-5t}e^{-(\sigma+j\omega)t}\,dt$

9.2. Consider the signal

$$x(t) = e^{5t}u(1-t),$$

and denote its Laplace transform by $X(s)$.

(a) Using eq. (9.3), evaluate $X(s)$ and specify its region of convergence.

(b) Determine the values of the finite numbers A and t_0 such that the Laplace transform $G(s)$ of

$$g(t) = Ae^{-5t}u(-t-t_0)$$

has the same algebraic form as $X(s)$. What is the region of convergence corresponding to $G(s)$?

9.3. Consider the signal

$$x(t) = e^{-5t}u(t) + e^{-\beta t}u(t),$$

and denote its Laplace transform by $X(s)$. What are the constraints placed on the real and imaginary parts of β if the region of convergence of $X(s)$ is $\Re e\{s\} > -3$?

9.4. For the Laplace transform of

$$x(t) = \begin{cases} e^t \sin 2t, & t \le 0 \\ 0, & t > 0 \end{cases},$$

indicate the location of its poles and its region of convergence.

9.5. For each of the following algebraic expressions for the Laplace transform of a signal, determine the number of zeros located in the finite s-plane and the number of zeros located at infinity:

(a) $-\dfrac{1}{s+1} + \dfrac{1}{s+3}$

(b) $\dfrac{s+1}{s^2-1}$

(c) $\dfrac{s^3-1}{s^2+s+1}$

9.6. An absolutely integrable signal $x(t)$ is known to have a pole at $s = 2$. Answer the following questions:
(a) Could $x(t)$ be of finite duration?
(b) Could $x(t)$ be left sided?
(c) Could $x(t)$ be right sided?
(d) Could $x(t)$ be two sided?

9.7. How many signals have a Laplace transform that may be expressed as

$$\frac{(s-1)}{(s+2)(s+3)(s^2+s+1)}$$

in its region of convergence?

9.8. Let $x(t)$ be a signal that has a rational Laplace transform with exactly two poles, located at $s = -1$ and $s = -3$. If $g(t) = e^{2t}x(t)$ and $G(j\omega)$ [the Fourier transform of $g(t)$] converges, determine whether $x(t)$ is left sided, right sided, or two sided.

9.9. Given that

$$e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a}, \qquad \mathcal{R}e\{s\} > \mathcal{R}e\{-a\},$$

determine the inverse Laplace transform of

$$X(s) = \frac{2(s+2)}{s^2+7s+12}, \qquad \mathcal{R}e\{s\} > -3.$$

9.10. Using geometric evaluation of the magnitude of the Fourier transform from the corresponding pole-zero plot, determine, for each of the following Laplace transforms, whether the magnitude of the corresponding Fourier transform is approximately lowpass, highpass, or bandpass:

(a) $H_1(s) = \dfrac{1}{(s+1)(s+3)}, \quad \mathcal{R}e\{s\} > -1$

(b) $H_2(s) = \dfrac{s}{s^2+s+1}, \quad \mathcal{R}e\{s\} > -\frac{1}{2}$

(c) $H_3(s) = \dfrac{s^2}{s^2+2s+1}, \quad \mathcal{R}e\{s\} > -1$

9.11. Use geometric evaluation from the pole-zero plot to determine the magnitude of the Fourier transform of the signal whose Laplace transform is specified as

$$X(s) = \frac{s^2-s+1}{s^2+s+1}, \qquad \mathcal{R}e\{s\} > -\frac{1}{2}.$$

9.12. Suppose we are given the following three facts about the signal $x(t)$:
1. $x(t) = 0$ for $t < 0$.
2. $x(k/80) = 0$ for $k = 1, 2, 3, \ldots$.
3. $x(1/160) = e^{-120}$.

Lat $X(s)$ denote the Laplace transform of $x(t)$, and determine which of the following statements is consistent with the given information about $x(t)$:
(a) $X(s)$ has only one pole in the finite s-plane.
(b) $X(s)$ has only two poles in the finite s-plane.
(c) $X(s)$ has more than two poles in the finite s-plane.

9.13. Let

$$g(t) = x(t) + \alpha x(-t),$$

where

$$x(t) = \beta e^{-t}u(t)$$

and the Laplace transform of $g(t)$ is

$$G(s) = \frac{8}{s^2 - 1}, \qquad -1 < \mathcal{R}e\{s\} < 1.$$

Determine the values of the constants α and β.

9.14. Suppose the following facts are given about the signal $x(t)$ with Laplace transform $X(s)$:

1. $x(t)$ is real and even.
2. $X(s)$ has four poles and no zeros in the finite s-plane.
3. $X(s)$ has a pole at $s = (1/2)e^{j\pi/4}$
4. $\int_{-\infty}^{\infty} x(t)\,dt = 4$.

Determine $X(s)$ and its ROC.

9.15. Consider two right-sided signals $x(t)$ and $y(t)$ related through the differential equations

$$\frac{dx(t)}{dt} = -2y(t) + \delta(t)$$

and

$$\frac{dy(t)}{dt} = 2x(t).$$

Determine $Y(s)$ and $X(s)$, along with their regions of convergence.

9.16. A causal LTI system S with impulse response $h(t)$ has its input $x(t)$ and output $y(t)$ related through a linear constant-coefficient differential equation of the form

$$\frac{d^3y(t)}{dt^3} + (1+\alpha)\frac{d^2y(t)}{dt^2} + \alpha(\alpha+1)\frac{dy(t)}{dt} + \alpha^2 y(t) = x(t).$$

(a) If

$$g(t) = \frac{dh(t)}{dt} + h(t),$$

how many poles does $G(s)$ have?

(b) For what real values of the parameter α is S guaranteed to be stable?

9.17. A causal LTI system S has the block diagram representation shown in Figure P9 17. Determine a differential equation relating the input $x(t)$ to the output $y(t)$ of this system.

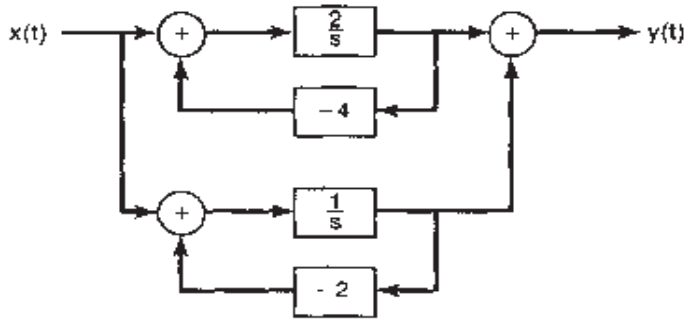


Figure P9.17

9.18. Consider the causal LTI system represented by the *RLC* circuit examined in Problem 3.20.

(a) Determine $H(s)$ and specify its region of convergence. Your answer should be consistent with the fact that the system is causal and stable.

(b) Using the pole-zero plot of $H(s)$ and geometric evaluation of the magnitude of the Fourier transform, determine whether the magnitude of the corresponding Fourier transform has an approximately lowpass, highpass, or bandpass characteristic.

(c) If the value of R is now changed to $10^{-3}\ \Omega$, determine $H(s)$ and specify its region of convergence.

(d) Using the pole-zero plot of $H(s)$ obtained in part (c) and geometric evaluation of the magnitude of the Fourier transform, determine whether the magnitude of the corresponding Fourier transform has an approximately lowpass, highpass, or bandpass characteristic.

9.19. Determine the unilateral Laplace transform of each of the following signals, and specify the corresponding regions of convergence:

(a) $x(t) = e^{2t}u(t+1)$

(b) $x(t) = \delta(t+1) + \delta(t) + e^{-2(t+3)}u(t+1)$

(c) $x(t) = e^{-2t}u(t) + e^{-4t}u(t)$

9.20. Consider the *RL* circuit of Problem 3.19.

(a) Determine the zero-state response of this circuit when the input current is $x(t) = e^{2t}u(t)$.

(b) Determine the zero-input response of the circuit for $t > 0^-$, given that

$$y(0^-) = 1.$$

(c) Determine the output of the circuit when the input current is $x(t) = e^{-2t}u(t)$ and the initial condition is the same as the one specified in part (b)

BASIC PROBLEMS

9.21. Determine the Laplace transform and the associated region of convergence and pole-zero plot for each of the following functions of time:

(a) $x(t) = e^{2t}u(t) + e^{-3t}u(t)$ **(b)** $x(t) = e^{-4t}u(t) + e^{-5t}(\sin 5t)u(t)$

(c) $x(t) = e^{2t}u(-t) + e^{3t}u(-t)$ **(d)** $x(t) = te^{-2|t|}$

(e) $x(t) = |t|e^{-2|t|}$ (f) $x(t) = |t|e^{2t}u(-t)$

(g) $x(t) = \begin{cases} 1, & 0 \le t \le 1 \\ 0, & \text{elsewhere} \end{cases}$ (h) $x(t) = \begin{cases} t, & 0 \le t \le 1 \\ 2-t, & 1 \le t \le 2 \end{cases}$

(i) $x(t) = \delta(t) + u(t)$ (j) $x(t) = \delta(3t) + u(3t)$

9.22. Determine the function of time, $x(t)$, for each of the following Laplace transforms and their associated regions of convergence:

(a) $\frac{1}{s^2+9}$, $\mathcal{R}e\{s\} > 0$

(b) $\frac{s}{s^2+9}$, $\mathcal{R}e\{s\} < 0$

(c) $\frac{s+1}{(s+1)^2+9}$, $\mathcal{R}e\{s\} < -1$

(d) $\frac{s+2}{s^2+7s+12}$, $-4 < \mathcal{R}e\{s\} < -3$

(e) $\frac{s+1}{s^2+5s+6}$, $-3 < \mathcal{R}e\{s\} < -2$

(f) $\frac{(s+1)^2}{s^2+s+1}$, $\mathcal{R}e\{s\} > \frac{1}{2}$

(g) $\frac{s^2-s+1}{(s+1)^2}$, $\mathcal{R}e\{s\} > -1$

9.23. For each of the following statements about $x(t)$, and for each of the four pole-zero plots in Figure P9.23, determine the corresponding constraint on the ROC:

1. $x(t)e^{-3t}$ is absolutely integrable.
2. $x(t) * (e^{-t}u(t))$ is absolutely integrable.
3. $x(t) = 0$, $t > 1$.
4. $x(t) = 0$, $t < -1$.

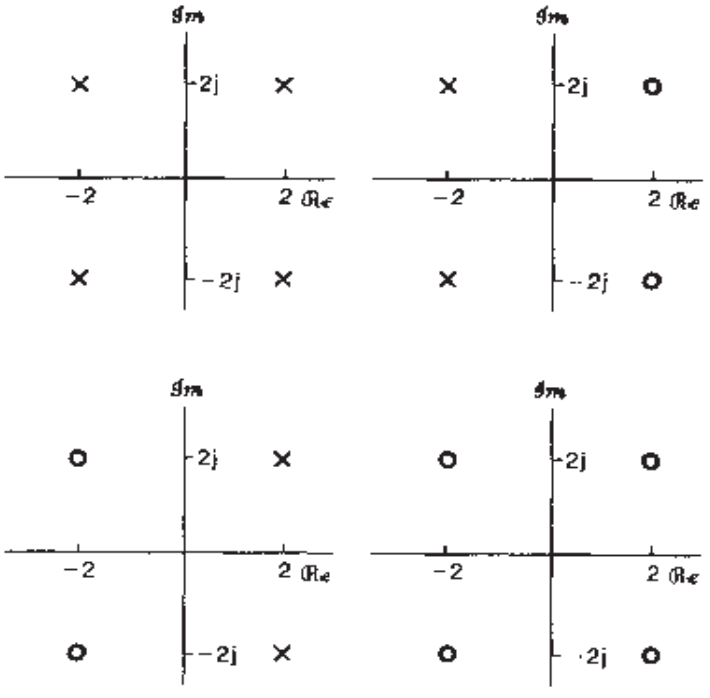


Figure P9.23

9.24. Throughout this problem, we will consider the region of convergence of the Laplace transforms always to include the $j\omega$-axis.

(a) Consider a signal $x(t)$ with Fourier transform $X(j\omega)$ and Laplace transform $X(s) = s + 1/2$. Draw the pole-zero plot for $X(s)$. Also, draw the vector whose length represents $|X(j\omega)|$ and whose angle with respect to the real axis represents $\measuredangle X(j\omega)$ for a given ω.

(b) By examining the pole-zero plot and vector diagram in part (a), determine a different Laplace transform $X_1(s)$ corresponding to the function of time, $x_1(t)$, so that

$$|X_1(j\omega)| = |X(j\omega)|,$$

but

$$x_1(t) \neq x(t).$$

Show the pole-zero plot and associated vectors that represent $X_1(j\omega)$.

(c) For your answer in part (b), determine, again by examining the related vector diagrams, the relationship between $\measuredangle X(j\omega)$ and $\measuredangle X_1(j\omega)$.

(d) Determine a Laplace transform $X_2(s)$ such that

$$\measuredangle X_2(j\omega) = \measuredangle X(j\omega),$$

but $x_2(t)$ is not proportional to $x(t)$. Show the pole-zero plot for $X_2(s)$ and the associated vectors that represent $X_2(j\omega)$.

(e) For your answer in part (d), determine the relationship between $|X_2(j\omega)|$ and $|X(j\omega)|$.

(f) Consider a signal $x(t)$ with Laplace transform $X(s)$ for which the pole-zero plot is shown in Figure P9.24. Determine $X_1(s)$ such that $|X(j\omega)| = |X_1(j\omega)|$ and all poles and zeros of $X_1(s)$ are in the left-half of the s-plane [i.e., $\mathcal{R}e\{s\} < 0$]. Also, determine $X_2(s)$ such that $\measuredangle X(j\omega) = \measuredangle X_2(j\omega)$ and all poles and zeros of $X_2(s)$ are in the left-half of the s-plane.

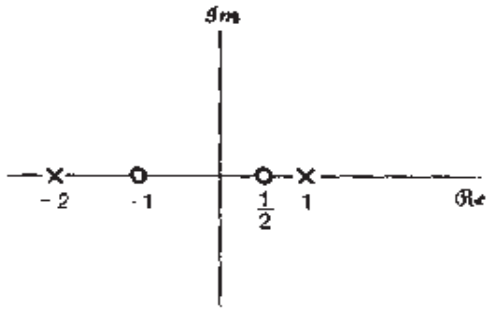


Figure P9.24

9.25. By considering the geometric determination of the Fourier transform, as developed in Section 9.4, sketch, for each of the pole-zero plots in Figure P9.25, the magnitude of the associated Fourier transform.

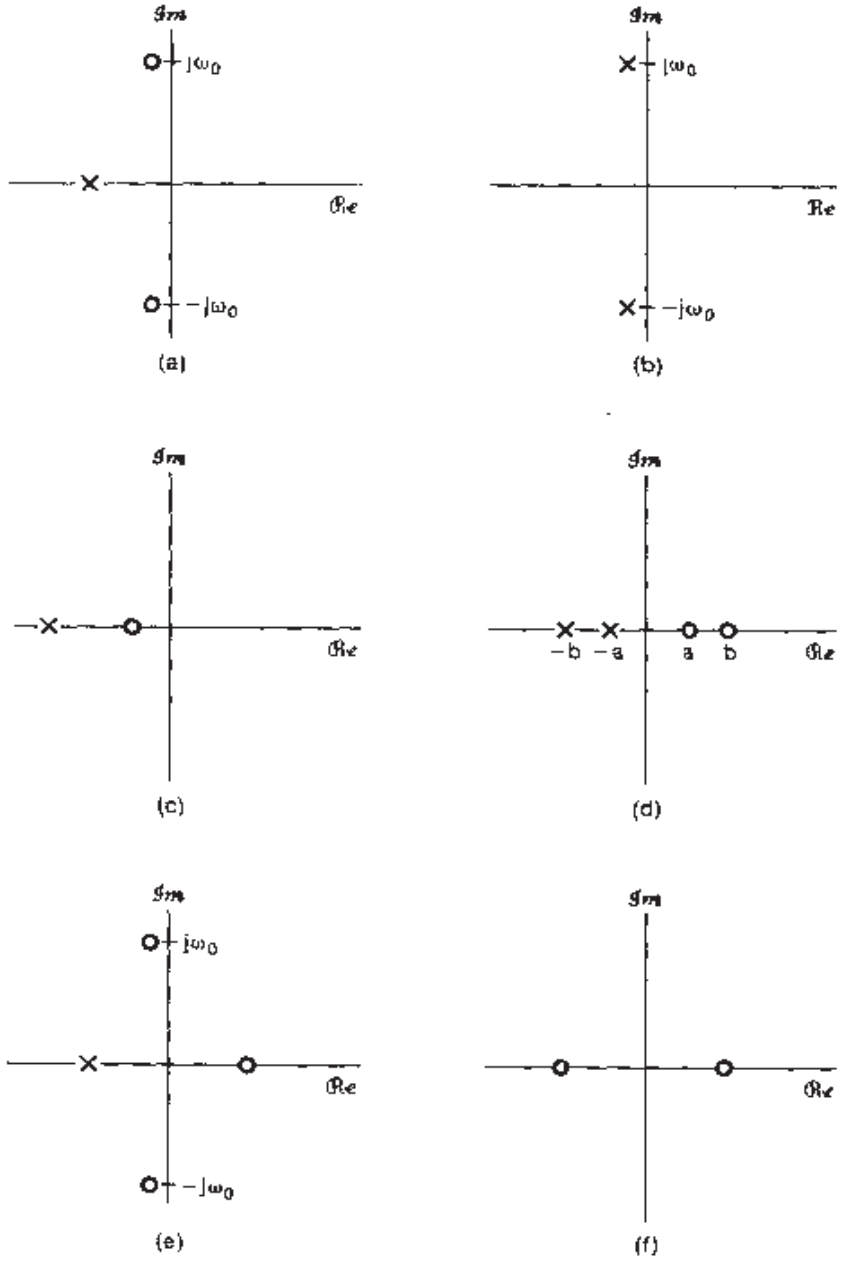


Figure P9.25

9.26. Consider a signal $y(t)$ which is related to two signals $x_1(t)$ and $x_2(t)$ by

$$y(t) = x_1(t-2) * x_2(-t+3)$$

where

$$x_1(t) = e^{-2t}u(t) \qquad \text{and} \qquad x_2(t) = e^{-3t}u(t).$$

Given that

$$e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a}, \quad \mathcal{R}e\{s\} > a,$$

use properties of the Laplace transform to determine the Laplace transform $Y(s)$ of $y(t)$.

9.27. We are given the following five facts about a real signal $x(t)$ with Laplace transform $X(s)$:

1. $X(s)$ has exactly two poles.
2. $X(s)$ has no zeros in the finite s-plane.
3. $X(s)$ has a pole at $s = -1 + j$.
4. $e^{2t}x(t)$ is not absolutely integrable.
5. $X(0) = 8$.

Determine $X(s)$ and specify its region of convergence.

9.28. Consider an LTI system for which the system function $H(s)$ has the pole-zero pattern shown in Figure P9.28.

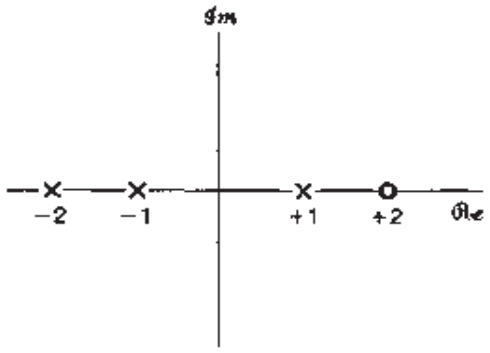


Figure P9.28

(a) Indicate all possible ROCs that can be associated with this pole-zero pattern.

(b) For each ROC identified in part (a), specify whether the associated system is stable and/or causal.

9.29. Consider an LTI system with input $x(t) = e^{-t}u(t)$ and impulse response $h(t) = e^{-2t}u(t)$.

(a) Determine the Laplace transforms of $x(t)$ and $h(t)$.

(b) Using the convolution property, determine the Laplace transform $Y(s)$ of the output $y(t)$.

(c) From the Laplace transform of $y(t)$ as obtained in part (b), determine $y(t)$.

(d) Verify your result in part (b) by explicitly convolving $x(t)$ and $h(t)$.

9.30. A pressure gauge that can be modeled as an LTI system has a time response to a unit step input given by $(1 - e^{-t} - te^{-t})u(t)$. For a certain input $x(t)$, the output is observed to be $(2 - 3e^{-t} + e^{-3t})u(t)$.

For this observed measurement, determine the true pressure input to the gauge as a function of time.

9.31. Consider a continuous-time LTI system for which the input $x(t)$ and output $y(t)$ are related by the differential equation

$$\frac{d^2y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t).$$

Let $X(s)$ and $Y(s)$ denote Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of $h(t)$, the system impulse response.

(a) Determine $H(s)$ as a ratio of two polynomials in s. Sketch the pole-zero pattern of $H(s)$.

(b) Determine $h(t)$ for each of the following cases:

1. The system is stable.
2. The system is causal.
3. The system is *neither* stable *nor* causal.

9.32. A causal LTI system with impulse response $h(t)$ has the following properties:

1. When the input to the system is $x(t) = e^{2t}$ for all t, the output is $y(t) = (1/6)e^{2t}$ for all t.
2. The impulse response $h(t)$ satisfies the differential equation

$$\frac{dh(t)}{dt} + 2h(t) = (e^{-4t})u(t) + bu(t),$$

where b is an unknown constant.

Determine the system function $H(s)$ of the system, consistent with the information above. There should be no unknown constants in your answer; that is, the constant b should *not* appear in the answer.

9.33. The system function of a causal LTI system is

$$H(s) = \frac{s+1}{s^2+2s+2}.$$

Determine and sketch the response $y(t)$ when the input is

$$x(t) = e^{|t|}, \qquad -\infty < t < \infty.$$

9.34. Suppose we are given the following information about a causal and stable LTI system S with impulse response $h(t)$ and a rational system function $H(s)$

1. $H(1) = 0.2$.
2. When the input is $u(t)$, the output is absolutely integrable.
3. When the input is $tu(t)$, the output is not absolutely integrable.
4. The signal $d^2h(t)/dt^2 + 2\,dh(t)/dt + 2h(t)$ is of finite duration.
5. $H(s)$ has exactly one zero at infinity.

Determine $H(s)$ and its region of convergence.

9.35. The input $x(t)$ and output $y(t)$ of a causal LTI system are related through the block-diagram representation shown in Figure P9.35.

(a) Determine a differential equation relating $y(t)$ and $x(t)$.

(b) Is this system stable?.

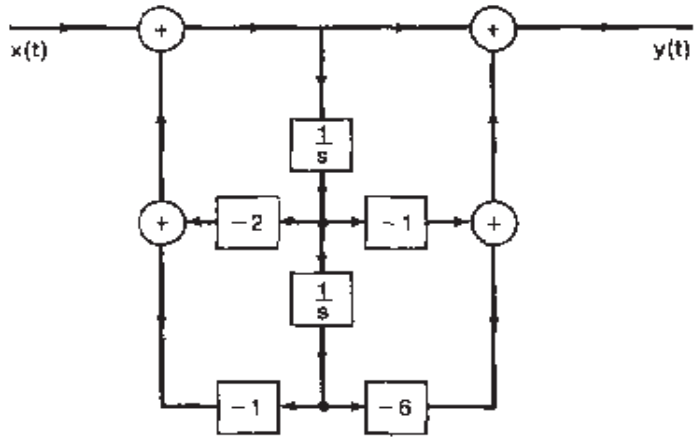


Figure P9.35

9.36. In this problem, we consider the construction of various types of block diagram representations for a causal LTI system S with input $x(t)$, output $y(t)$, and system function

$$H(s) = \frac{2s^2 + 4s - 6}{s^2 + 3s + 2}.$$

To derive the direct-form block diagram representation of S, we first consider a causal LTI system S_1 that has the same input $x(t)$ as S, but whose system function is

$$H_1(s) = \frac{1}{s^2 + 3s + 2}.$$

With the output of S_1 denoted by $y_1(t)$, the direct-form block diagram representation of S_1 is shown in Figure P9.36. The signals $e(t)$ and $f(t)$ indicated in the figure represent respective inputs into the two integrators.

(a) Express $y(t)$ (the output of S) as a linear combination of $y_1(t)$, $dy_1(t)/dt$, and $d^2y_1(t)/dt^2$

(b) How is $dy_1(t)/dt$ related to $f(t)$?

(c) How is $d^2y_1(t)/dt^2$ related to $e(t)$?

(d) Express $y(t)$ as a linear combination of $e(t)$, $f(t)$, and $y_1(t)$.

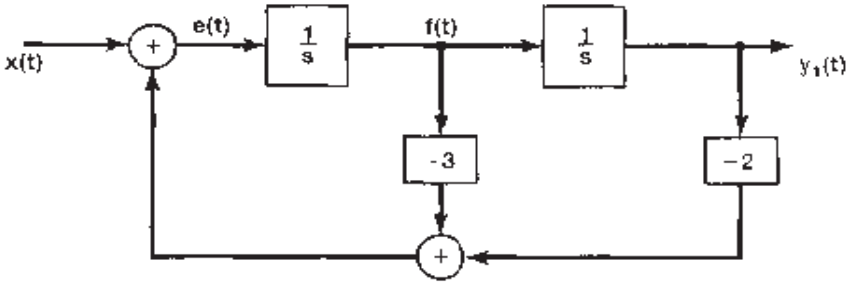


Figure P9.36

(e) Use the result from the previous part to extend the direct-form block diagram representation of S_1 and create a block diagram representation of S.

(f) Observing that

$$H(s) = \left(\frac{2(s-1)}{s+2}\right)\left(\frac{s+3}{s+1}\right),$$

draw a block diagram representation for S as a cascade combination of two subsystems.

(g) Observing that

$$H(s) = 2 + \frac{6}{s+2} - \frac{8}{s+1},$$

draw a block-diagram representation for S as a parallel combination of three subsystems.

9.37. Draw a direct-form representation for the causal LTI systems with the following system functions:

(a) $H_1(s) = \frac{s+1}{s^2+5s+6}$ (b) $H_2(s) = \frac{s^2-5s+6}{s^2+7s+10}$ (c) $H_3(s) = \frac{s}{(s+2)^2}$

9.38. Consider a fourth-order causal LTI system S whose system function is specified as

$$H(s) = \frac{1}{(s^2 - s + 1)(s^2 + 2s + 1)}.$$

(a) Show that a block diagram representation for S consisting of a cascade of four first-order sections will contain multiplications by coefficients that are not purely real.

(b) Draw a block diagram representation for S as a *cascade* interconnection of two second-order systems, each of which is represented in direct form. There should be no multiplications by nonreal coefficients in the resulting block diagram.

(c) Draw a block diagram representation for S as a *parallel* interconnection of two second-order systems, each of which is represented in direct form. There should be no multiplications by nonreal coefficients in the resulting block diagram.

9.39. Let

$$x_1(t) = e^{-2t}u(t) \quad \text{and} \quad x_2(t) = e^{-3(t+1)}u(t+1).$$

(a) Determine the unilateral Laplace transform $\mathfrak{X}_1(s)$ and the bilateral Laplace transform $X_1(s)$ for the signal $x_1(t)$.

(b) Determine the unilateral Laplace transform $\mathfrak{X}_2(s)$ and the bilateral Laplace transform $X_2(s)$ for the signal $x_2(t)$.

(c) Take the inverse bilateral Laplace transform of the product $X_1(s)X_2(s)$ to determine the signal $g(t) = x_1(t) * x_2(t)$.

(d) Show that the inverse unilateral Laplace transform of the product $\mathfrak{X}_1(s)\mathfrak{X}_2(s)$ is not the same as $g(t)$ for $t > 0^-$.

9.40. Consider the system S characterized by the differential equation

$$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = x(t).$$

(a) Determine the zero-state response of this system for the input $x(t) = e^{-4t}u(t)$.
(b) Determine the zero-input response of the system for $t > 0^-$, given that

$$y(0^-) = 1, \qquad \left.\frac{dy(t)}{dt}\right|_{t=0^-} = -1, \qquad \left.\frac{d^2y(t)}{dt^2}\right|_{t=0} = 1.$$

(c) Determine the output of S when the input is $x(t) = e^{4t}u(t)$ and the initial conditions are the same as those specified in part (b).

ADVANCED PROBLEMS

9.41. **(a)** Show that, if $x(t)$ is an even function, so that $x(t) = x(-t)$, then $X(s) = X(-s)$.
(b) Show that, if $x(t)$ is an odd function, so that $x(t) = -x(-t)$, then $X(s) = -X(-s)$.
(c) Determine which, if any, of the pole-zero plots in Figure P9.41 could correspond to an even function of time. For those that could, indicate the required ROC.

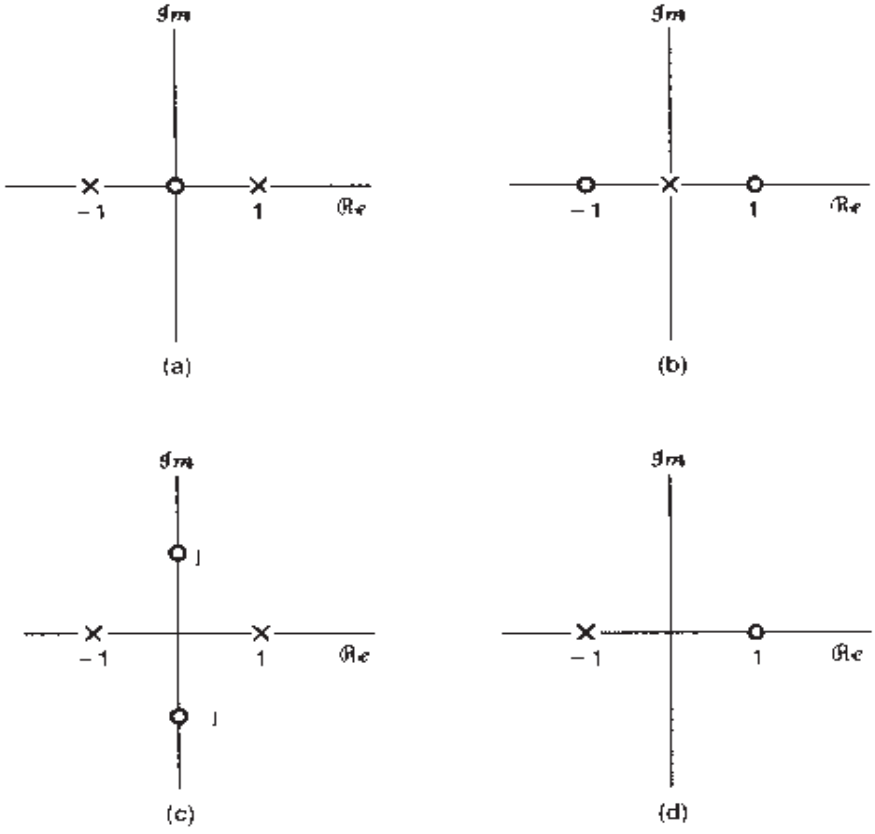


Figure P9.41

9.42. Determine whether each of the following statements is true or false. If a statement is true, construct a convincing argument for it. If it is false, give a counterexample.
(a) The Laplace transform of $t^2u(t)$ does not converge anywhere on the s-plane.
(b) The Laplace transform of $e^{t^2}u(t)$ does not converge anywhere on the s-plane.
(c) The Laplace transform of $e^{j\omega_0 t}$ does not converge anywhere on the s-plane.
(d) The Laplace transform of $e^{j\omega_0 t}u(t)$ does not converge anywhere on the s-plane.
(e) The Laplace transform of $|t|$ does not converge anywhere on the s-plane.

9.43. Let $h(t)$ be the impulse response of a causal and stable LTI system with a rational system function.
(a) Is the system with impulse response $dh(t)/dt$ guaranteed to he causal and stable?
(b) Is the system with impulse response $\int_{-\infty}^{t} h(\tau)\,d\tau$ guaranteed to he causal and unstable?

9.44. Let $x(t)$ be the sampled sigual specified as

$$x(t) = \sum_{n=0}^{\infty} e^{-nT}\,\delta(t - nT),$$

where $T > 0$.
(a) Determine $X(s)$, including its region of convergence.
(b) Sketeh the pole-zero plot for $X(s)$.
(c) Use geometric interpretation of the pole-zero plot to argue that $X(j\omega)$ is periodic.

9.45. Consider the LTI system shown in Figure P9.45(a) for which we are given the following information:

$$X(s) = \frac{s+2}{s-2},$$

$$x(t) = 0, \qquad t > 0,$$

and

$$y(t) = -\frac{2}{3}e^{2t}u(-t) + \frac{1}{3}e^{-t}u(t). \qquad \text{[See Figure P9.45(b).]}$$

(a) Determine $H(s)$ and its region of convergence.
(b) Determine $h(t)$.

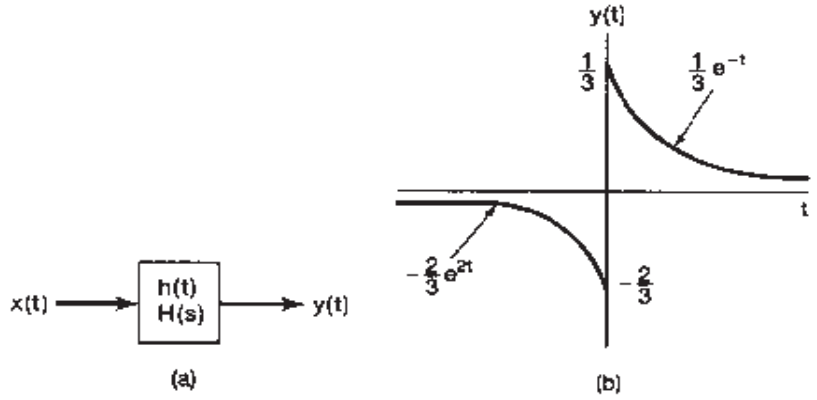


Figure P9.45

(c) Using the system function $H(s)$ found in part (a), determine the output $y(t)$ if the input is

$$x(t) = e^{3t}, \qquad -\infty < t < +\infty.$$

9.46. Let $H(s)$ represent the system function for a causal, stable system. The input to the system consists of the sum of three terms, one of which is an impulse $\delta(t)$ and another a complex exponential of the form $e^{s_0 t}$, where s_0 is a complex constant. The output is

$$y(t) = -6e^{-t}u(t) + \frac{4}{34}e^{4t}\cos 3t + \frac{18}{34}e^{4t}\sin 3t + \delta(t).$$

Determine $H(s)$, consistently with this information.

9.47. The signal

$$y(t) = e^{-2t}u(t)$$

is the output of a causal all-pass system for which the system function is

$$H(s) = \frac{s-1}{s+1}.$$

(a) Find and sketch at least two possible inputs $x(t)$ that could produce $y(t)$.

(b) What is the input $x(t)$ if it is known that

$$\int_{-\infty}^{\infty} |x(t)|\,dt < \infty?$$

(c) What is the input $x(t)$ if it is known that a stable (but not necessarily causal) system exists that will have $x(t)$ as an output if $y(t)$ is the input? Find the impulse response $h(t)$ of this filter, and show by direct convolution that it has the property claimed [i.e., that $y(t) * h(t) = x(t)$].

9.48. The *inverse* of an LTI system $H(s)$ is defined as a system that, when cascaded with $H(s)$, results in an overall transfer function of unity or, equivalently, an overall impulse response that is an impulse.

(a) If $H_1(s)$ denotes the transfer function of an inverse system for $H(s)$, determine the general algebraic relationship between $H(s)$ and $H_1(s)$.

(b) Shown in Figure P9.48 is the pole-zero plot for a stable, causal system $H(s)$. Determine the pole-zero plot for the associated inverse system.

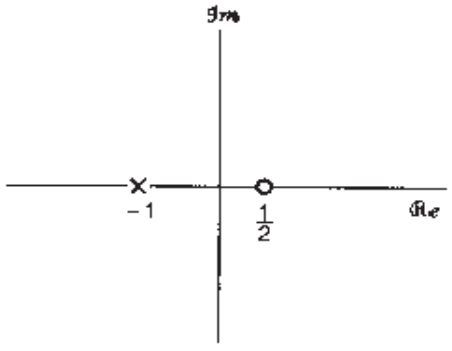


Figure P9.48

9.49. A class of systems, referred to as minimum-delay or minimum-phase systems, is sometimes defined through the statement that these systems are causal and stable and that the inverse systems are also causal and stable.

Based on the preceding definition, develop an argument to demonstrate that all poles *and* zeros of the transfer function of a minimum-delay system must be in the left half of the s-plane [i.e., $\mathcal{R}e\{s\} < 0$].

9.50. Determine whether or not each of the following statements about LTI systems is true. If a statement is true, construct a convincing argument for it. If it is false, give a counterexample.

(a) A stable continuous-time system must have all its poles in the left half of the s-plane [i.e., $\mathcal{R}e\{s\} < 0$].

(b) If a system function has more poles than zeros, and the system is causal, the step response will be continuous at $t = 0$.

(c) If a system function has more poles than zeros, and the system is not restricted to be causal, the step response can be discontinuous at $t = 0$.

(d) A stable, causal system must have all its poles and zeros in the left half of the s-plane.

9.51. Consider a stable and causal system with a real impulse response $h(t)$ and system function $H(s)$. It is known that $H(s)$ is rational, one of its poles is at $-1 + j$, one of its zeros is at $3 + j$, and it has exactly two zeros at infinity. For each of the following statements, determine whether it is true, whether it is false, or whether there is insufficient information to determine the statement's truth.

(a) $h(t)e^{-3t}$ is absolutely integrable.

(b) The ROC for $H(s)$ is $\mathcal{R}e\{s\} > -1$.

(c) The differential equation relating inputs $x(t)$ and outputs $y(t)$ for S may be written in a form having only real coefficients.

(d) $\lim_{s\to\infty} H(s) = 1$.

(e) $H(s)$ does not have fewer than four poles.

(f) $H(j\omega) = 0$ for at least one finite value of ω.

(g) If the input to S is $e^{3t}\sin t$, the output is $e^{3t}\cos t$.

9.52. As indicated in Section 9.5, many of the properties of the Laplace transform and their derivation are analogous to corresponding properties of the Fourier transform and their derivation, as developed in Chapter 4. In this problem, you are asked to outline the derivation of a number of the Laplace transform properties.

Observing the derivation for the corresponding property in Chapter 4 for the Fourier transform, derive each of the following Laplace transform properties. Your derivation must include a consideration of the region of convergence.

(a) Time shifting (Section 9.5.2)

(b) Shifting in the s-domain (Section 9.5.3)

(c) Time scaling (Section 9.5.4)

(d) Convolution property (Section 9.5.6)

9.53. As presented in Section 9.5.10, the initial-value theorem states that, for a signal $x(t)$ with Laplace transform $X(s)$ and for which $x(t) = 0$ for $t < 0$, the initial value of $x(t)$ [i.e., $x(0+)$] can be obtained from $X(s)$ through the relation

$$x(0+) = \lim_{s\to\infty} sX(s). \qquad \text{[eq. (9.110)]}$$

First, we note that, since $x(t) = 0$ for $t < 0$, $x(t) = x(t)u(t)$. Next, expanding $x(t)$ as a Taylor series at $t = 0+$, we obtain

$$x(t) = \left[x(0+) + x^{(1)}(0+)t + \cdots + x^{(n)}(0+)\frac{t^n}{n!} + \cdots\right]u(t), \tag{P9.53–1}$$

where $x^{(n)}(0+)$ denotes the nth derivative of $x(t)$ evaluated at $t = 0+$.

(a) Determine the Laplace transform of an arbitrary term $x^{(n)}(0+)(t^n/n!)u(t)$ on the right-hand side of eq. (P9.53–1). (You may find it helpful to review Example 9.14.)

(b) From your result in part (a) and the expansion in eq. (P9.53–1), show that $X(s)$ can be expressed as

$$X(s) = \sum_{n=0}^{\infty} x^{(n)}(0+)\frac{1}{s^{n+1}}.$$

(c) Demonstrate that eq. (9.110) follows from the result of part (b)

(d) By first determining $x(t)$, verify the initial-value theorem for each of the following examples:

(1) $X(s) = \frac{1}{s+2}$

(2) $X(s) = \frac{s+1}{(s+2)(s+3)}$

(e) A more general form of the initial-value theorem states that if $x^{(n)}(0+) = 0$ for $n < N$, then $x^{(N)}(0+) = \lim_{s\to\infty} s^{N+1}X(s)$. Demonstrate that this more general statement also follows from the result in part (b).

9.54. Consider a real-valued signal $x(t)$ with Laplace transform $X(s)$.

(a) By applying complex conjugation to both sides of eq. (9.56), show that $X(s) = X^*(s^*)$.

(b) From your result in (a), show that if $X(s)$ has a pole (zero) at $s = s_0$, it must also have a pole (zero) at $s = s_0^*$; i.e., for $x(t)$ real, the poles and zeros of $X(s)$ that are not on the real axis must occur in complex conjugate pairs.

9.55. In Section 9.6, Table 9.2, we listed a number of Laplace transform pairs, and we indicated specifically how transform pairs 1 through 9 follow from Examples 9.1 and 9.14, together with various properties from Table 9.1.

By exploiting properties from Table 9.1, show how transform pairs 10 through 16 follow from transform pairs 1 through 9 in Table 9.2.

9.56. The Laplace transform is said to exist for a specific complex s if the magnitude of the transform is finite—that is, if $|X(s)| < \infty$.

Show that a *sufficient* condition for the existence of the transform $X(s)$ at $s = s_0 = \sigma_0 + j\omega_0$ is that

$$\int_{-\infty}^{+\infty} |x(t)\, e^{-\sigma_0 t}\, dt < \infty.$$

In other words, show that $x(t)$ exponentially weighted by $e^{-\sigma_0 t}$ is absolutely integrable. You will need to use the result that, for a complex function $f(t)$,

$$\left|\int_a^b f(t)\,dt\right| \leq \int_a^b |f(t)|\,dt. \tag{P9.56-1}$$

Without rigorously proving eq. (P9.56-1), argue its plausibility.

9.57. The Laplace transform $X(s)$ of a signal $x(t)$ has four poles and an unknown number of zeros. The signal $x(t)$ is known to have an impulse at $t = 0$. Determine what information, if any, this provides about the number of zeros and their locations.

9.58. Let $h(t)$ be the impulse response of a causal and stable LTI system with rational system function $H(s)$. Show that $g(t) = \mathcal{R}e\{h(t)\}$ is also the impulse response of a causal and stable system.

9.59. If $\mathfrak{X}(s)$ denotes the unilateral Laplace transform of $x(t)$, determine, in terms of $\mathfrak{X}(s)$, the unilateral Laplace transform of:

(a) $x(t-1)$ **(b)** $x(t+1)$

(c) $\int_{-\infty}^{t} x(\tau)\,d\tau$ **(d)** $\frac{d^3x(t)}{dt^3}$

EXTENSION PROBLEMS

9.60. In long-distance telephone communication, an echo is sometimes encountered due to the transmitted signal being reflected at the receiver, sent back down the line, reflected again at the transmitter, and returned to the receiver. The impulse response for a system that models this effect is shown in Figure P9.60, where we have assumed that only one echo is received. The parameter T corresponds to the one-way travel time along the communication channel, and the parameter α represents the attenuation in amplitude between transmitter and receiver.

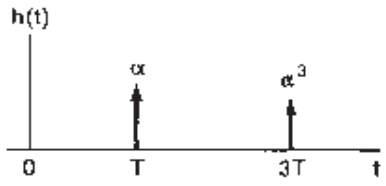


Figure P9.60

(a) Determine the system function $H(s)$ and associated region of convergence for the system.

(b) From your result in part (a), you should observe that $H(s)$ does not consist of a ratio of polynomials. Nevertheless, it is useful to represent it in terms of poles and zeros, where, as usual, the zeros are the values of s for which $H(s) = 0$

and the poles are the values of s for which $1/H(s) = 0$. For the system function found in part (a), determine the zeros and demonstrate that there are no poles.

(c) From your result in part (b), sketch the pole-zero plot for $H(s)$

(d) By considering the appropriate vectors in the s-plane, sketch the magnitude of the frequency response of the system.

9.61. The autocorrelation function of a signal $x(t)$ is defined as

$$\phi_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)\,dt.$$

(a) Determine, in terms of $x(t)$, the impulse response $h(t)$ of an LTI system for which, when the input is $x(t)$, the output is $\phi_{xx}(t)$ [Figure P9.61(a)].

(b) From your answer in part (a), determine $\Phi_{xx}(s)$, the Laplace transform of $\phi_{xx}(\tau)$ in terms of $X(s)$. Also, express $\Phi_{xx}(j\omega)$, the Fourier transform of $\phi_{xx}(\tau)$, in terms of $X(j\omega)$.

(c) If $x(t)$ has the pole-zero pattern and ROC shown in Figure P9.61(b), sketch the pole-zero pattern and indicate the ROC for $\phi_{xx}(\tau)$.

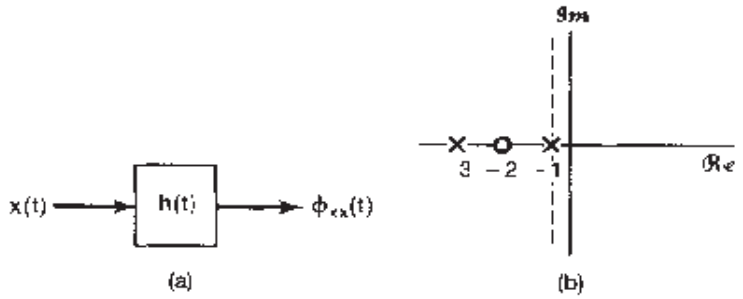


Figure P9.61

9.62. In a number of applications in signal design and analysis, the class of signals

$$\phi_n(t) = e^{-t/2}L_n(t)u(t), \quad n = 0, 1, 2, \ldots, \tag{P9.62-1}$$

where

$$L_n(t) = \frac{e^t}{n!}\frac{d^n}{dt^n}(t^n e^{-t}), \tag{P9.62-2}$$

is encountered.

(a) The functions $L_n(t)$ are referred to as *Laguerre polynomials*. To verify that they in fact have the form of polynomials, determine $L_0(t)$, $L_1(t)$, and $L_2(t)$ explicitly.

(b) Using the properties of the Laplace transform in Table 9.1 and Laplace transform pairs in Table 9.2, determine the Laplace transform $\Phi_n(s)$ of $\phi_n(t)$.

(c) The set of signals $\phi_n(t)$ can be generated by exciting a network of the form in Figure P9.62 with an impulse. From your result in part (b), determine $H_1(s)$ and $H_2(s)$ so that the impulse responses along the cascade chain are the signals $\phi_n(t)$ as indicated.

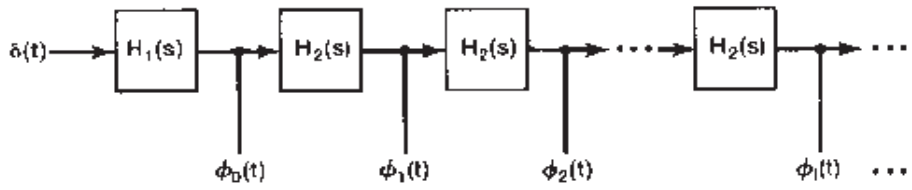


Figure P9.62

9.63. In filter design, it is often possible and convenient to transform a lowpass filter to a highpass filter and vice versa. With $H(s)$ denoting the transfer function of the original filter and $G(s)$ that of the transformed filter, one such commonly used transformation consists of replacing s by $1/s$; that is,

$$G(s) = H\left(\frac{1}{s}\right).$$

(a) For $H(s) = 1/(s + 1/2)$, sketch $|H(j\omega)|$ and $G(j\omega)|$.

(b) Determine the linear constant-coefficient differential equation associated with $H(s)$ and with $G(s)$.

(c) Now consider a more general case in which $H(s)$ is the transfer function associated with the linear constant-coefficient differential equation in the general form

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{N} b_k \frac{d^k x(t)}{dt^k}. \qquad \text{(P9.63–1)}$$

Without any loss of generality, we have assumed that the number of derivatives N is the same on both sides of the equation, although in any particular case, some of the coefficients may be zero. Determine $H(s)$ and $G(s)$.

(d) From your result in part (c), determine, in terms of the coefficients in eq. (P9.63–1), the linear constant-coefficient differential equation associated with $G(s)$.

9.64. Consider the *RLC* circuit shown in Figure 9.27 with input $x(t)$ and output $y(t)$.

(a) Show that if R, L, and C are all positive, then this LTI system is stable.

(b) How should R, L, and C be related to each other so that the system represents a second-order Butterworth filter?

9.65. **(a)** Determine the differential equation relating $v_i(t)$ and $v_0(t)$ for the *RLC* circuit of Figure P9.65.

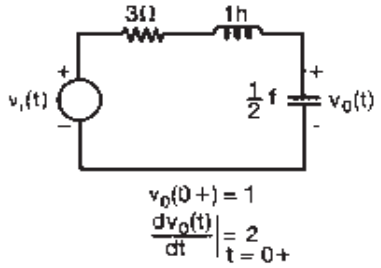


Figure P9.65

(b) Suppose that $v_i(t) = e^{-3t}u(t)$. Using the unilateral Laplace transform, determine $v_o(t)$ for $t > 0$

9.66. Consider the RL circuit shown in Figure P9.66. Assume that the current $i(t)$ has reached a steady state with the switch at position A. At time $t = 0$, the switch is moved from position A to position B.

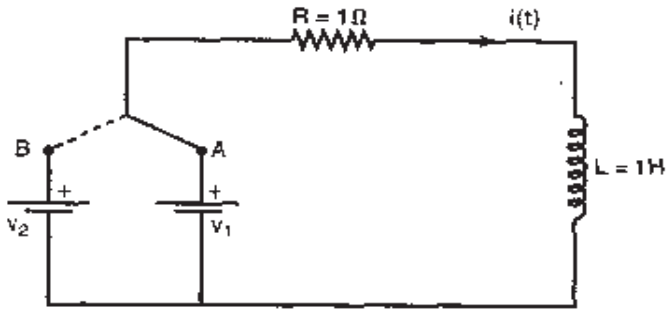


Figure P9.66

(a) Find the differential equation relating $i(t)$ and v_2 for $t > 0^-$. Specify the initial condition (i.e., the value of $i(0^-)$) for this differential equation in terms of v_1.

(b) Using the properties of the unilateral Laplace transform in Table 9.3, determine and plot the current $i(t)$ for each of the following values of v_1 and v_2:

(i) $v_1 = 0\ V, v_2 = 2\ V$

(ii) $v_1 = 4\ V, v_2 = 0\ V$

(iii) $v_1 = 4\ V, v_2 = 2\ V$

Using your answers for (i), (ii), and (iii), argue that the current $i(t)$ may be expressed as a sum of the circuit's zero-state response and zero-input response.